

```
1 2026-03-04 16:12:58] Connected. Run ID: 69a823598d0e1d2c5401271f
2 [2026-03-04 16:12:58] Configuration:
3 [2026-03-04 16:12:58] CDM Schema: clnprw_optum
4 [2026-03-04 16:12:58] Work Schema: osk02156
5 [2026-03-04 16:12:58] Input Cohort: ELIG_COH_FINAL
6 [2026-03-04 16:12:58] Induction Window: 60 days
7 [2026-03-04 16:12:58] MAP Discon Gap: 90 days
8 [2026-03-04 16:12:58] Medical Day Supply: 28 days
9 [2026-03-04 16:12:58] LOT Discon Gap: 90 days
10 [2026-03-04 16:12:58] Loaded codelist from CSV: /mnt/code/codelist/cl_mma_rollup.csv ( 30 rows)
11 [2026-03-04 16:12:58] Loaded codelist from CSV: /mnt/code/codelist/cl_mma_codelist.csv ( 528 rows)
12 [2026-03-04 16:12:58] Loaded codelist from CSV: /mnt/code/codelist/permisible_subs.csv ( 3 rows)
13 [2026-03-04 16:12:58] Loaded codelist from CSV: /mnt/code/codelist/cl_sct_codelist.csv ( 128 rows)
14 [2026-03-04 16:12:58]
15 [2026-03-04 16:12:58] STEP S00_mma_rollup
16 [2026-03-04 16:12:58]
17 [2026-03-04 16:12:58] Completed in 0.3 s
18 [2026-03-04 16:12:58] n_rows n_meds n_monomaint n_conditioning n_other_cancer
19 1 30 26 5 10
20 [2026-03-04 16:14:20]
21 [2026-03-04 16:14:20] STEP S01_mma_codelist
22 [2026-03-04 16:14:20]
23 [2026-03-04 16:14:21] Completed in 0.4 s
24 [2026-03-04 16:14:21] n_rows n_meds n_code_types
25 1 528 26 2
26 [2026-03-04 16:14:22]
27 [2026-03-04 16:14:22] STEP S02_permisible_subs
28 [2026-03-04 16:14:22]
29 [2026-03-04 16:14:22] Completed in 0.2 s
30 [2026-03-04 16:14:22] n_rows n_orig_meds
31 1 3 3
32 [2026-03-04 16:14:22] Checking codelist <-> rollup consistency...
33 [2026-03-04 16:14:23] OK: All codelist meds found in rollup.
34 [2026-03-04 16:14:24] OK: All rollup meds have at least one code in codelist.
```

```
new 2
new 1
new 3

34 [2026-03-04 16:14:24] OK: All rolup meds have at least one code in codelist.
35 [2026-03-04 16:14:25] Code type distribution in codelist:
36 CL_CODE_TYPE n_codes
37 1 HCPCS 67
38 2 NDC 461
39 [2026-03-04 16:14:26] OK: Each MED_ABBR maps to exactly one class.
40 [2026-03-04 16:14:28] Rolup meds: BELA, BEND, BORT, CARF, CILT, CISP, CYCL, DARA, DOPL, DOXO, ELIOT, ELRA, ETOP, IDEC,
41 ISAT, IXAZ, LENA, LINV, MELP, PANO, POMA, SELI, TALQ, TECH, THAL, VENE
42 [2026-03-04 16:14:28] Rolup classes: ABCMA, ACD38, ASLAME7, ATCELL, BLC2I, HIST, IMMUNOMOD, MUSTARD, NUCLEAR, PLAT,
43 PROTNHIB, TOPOINHIB
44 [2026-03-04 16:14:28] STEP S03_patient_input
45 [2026-03-04 16:14:28] Completed in 0.5 s
46 n_patients min_index max_obs end_n_disenrolled_before_enddate
47 1 8598 2016-01-04 2025-06-30 2610
48 [2026-03-04 16:14:29] STEP S04_mma_med_raw
49 [2026-03-04 16:14:29] Completed in 0.7 s
50 [2026-03-04 16:14:29] n_rows n_patients n_meds n_pharmacy_rows n_medical_rows n_from_proc cd n_from_bill_proc n_from_med ndc
51 [2026-03-04 16:14:30] 1 772682 8528 25 96870 675812 438123 72603 165086
52 n_from_med_procedure n_from_rx_ndc
53 1 0 96870
54 [2026-03-04 16:15:10] STEP S05_mma_med_processed
55 [2026-03-04 16:15:10] Completed in 1.9 s
56 [2026-03-04 16:15:10] n_rows n_pharmacy_rows n_medical_rows min_day_supply max_day_supply
57 [2026-03-04 16:15:10] 1 358647 96661 261986 1 100
58 [2026-03-04 16:15:48] STEP S06_map_med
59 [2026-03-04 16:15:48] Completed in 0.8 s
60 [2026-03-04 16:15:48]
61
62
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65 [2026-03-04 16:15:48] Completed in 0.8 s
66 n_maps n_patients n_meds avg_map len_days n_discontinuations
67 1 95650 8528 25 61.34352 14627
68 [2026-03-04 16:16:17] STEP S07_map_stacked
69 [2026-03-04 16:16:17]
70 [2026-03-04 16:16:17] Completed in 0.9 s
71 [2026-03-04 16:16:18]
72 n_rows
73 1 95650
74 [2026-03-04 16:16:45] STEP S08_lotl_start
75 [2026-03-04 16:16:45]
76 [2026-03-04 16:16:45] Completed in 0.9 s
77 [2026-03-04 16:16:45]
78 n_patients_with_lotl min_lotl_start max_lotl_start
79 1 8528 2016-01-05 2025-06-30
80 [2026-03-04 16:17:13] STEP S09_lotl_induction_meds
81 [2026-03-04 16:17:13]
82 [2026-03-04 16:17:13] Completed in 1.3 s
83 [2026-03-04 16:17:14]
84 n_rows n_patients avg_induction_meds
85 1 8528 1.898687
86 [2026-03-04 16:17:42] STEP S10_lotl_base
87 [2026-03-04 16:17:42]
88 [2026-03-04 16:17:42] Completed in 4.3 s
89 [2026-03-04 16:17:46] n_patients avg_induction_meds avg_base_length n_with_discon dt n_with_add med
90 1 8528 1.898687 456.2432 3363 2291
91 [2026-03-04 16:18:58] STEP S11_sct_codelist
92 [2026-03-04 16:18:58]
93 [2026-03-04 16:18:58] Completed in 0.3 s
94 [2026-03-04 16:18:58]
95 [2026-03-04 16:18:58]
96 SCT_TYPE n_codes
97 1 ALLOGENIC 79
98 2 AUTOLOGOUS 29
```

```

98 2 AUTOLOGOUS 29
99 3 UNKNOWN 20
100 [2026-03-04 16:18:59]
101 [2026-03-04 16:18:59] STEP S12_sct_claims_raw
102 [2026-03-04 16:18:59]
103 [2026-03-04 16:18:59] Completed in 0.8 s
104 [1] SCT_TYPE n_claims n_patients min_date max_date
105 <0 rows> (or 0-length row.names)
106 [2026-03-04 16:19:00]
107 [2026-03-04 16:19:00] STEP S13_tx_auto_dates
108 [2026-03-04 16:19:00]
109 [2026-03-04 16:19:01] Completed in 0.7 s
110 n_auto_tx_events n_patients min_seq max_seq
111 1 0 NA NA
112 [2026-03-04 16:19:02]
113 [2026-03-04 16:19:02] STEP S14_tx_allo_cart_dates
114 [2026-03-04 16:19:02]
115 [2026-03-04 16:19:03] Completed in 0.7 s
116 [1] SCT_TYPE n_events n_patients
117 <0 rows> (or 0-length row.names)
118 [2026-03-04 16:19:04]
119 [2026-03-04 16:19:04] STEP S15_lot1_sct
120 [2026-03-04 16:19:04]
121 [2026-03-04 16:19:14] Completed in 10.1 s
122 n_patients n_with_auto n_with_allo n_with_cart n_tandem n_single_auto n_with_sct_end
123 1 8528 0 0 0 0
124 [2026-03-04 16:21:07]
125 [2026-03-04 16:21:07] STEP S16_lot1_base_end
126 [2026-03-04 16:21:07]
127 [2026-03-04 16:21:27] Completed in 19.2 s
128 LOT1_BASE_END_REASON n
129 1 CENSORED 3970
130 2 DISCONTINUATION 2267
131 3 MED_ADD 2291

```

Normal text file

length: 16,347 lines: 270

Ln: 98 Col: 21 Pos: 6,482

Search

lpy

MED_ADD 2291

3 [2026-03-04 16:24:04] NOTE: Maintenance (mono/dual) specs not yet provided; LOT1_BASE_END_REASON

132 [2026-03-04 16:24:04] does not yet include MAINTENANCE_START. Will need integration when available.

133 [2026-03-04 16:24:04] Running NDC format QC...

134 [2026-03-04 16:24:05] NDC length distribution in codelist:

135 ndc_len n

136 11 461

137 [2026-03-04 16:24:06] NDC length distribution in RX claims:

138 ndc_len n

139 1 3 144

140 2 4 16

141 3 11 910553

142 [2026-03-04 16:24:06] Running validation QC suite...

143 [2026-03-04 16:24:06] [A] MMA_MED extraction coverage:

144 CODE_TYPE CLAIM_TYPE n_claims n_patients n_meds

145 HCPCS medical 260556 7708 17

146 NDC medical 1430 206 11

147 NDC pharmacy 96661 6384 19

148 [2026-03-04 16:24:30] [B] MAP algorithm spot checks:

149 [2026-03-04 16:24:52] MAPS with END < START: 0 (OK)

150 [2026-03-04 16:25:14] MAPS where END_DT != max(rx_runout, med_runout): 0 (OK)

151 [2026-03-04 16:25:35] MAPS with both pharmacy + medical sources: 321

152 [2026-03-04 16:25:35] [C] OBS_END_DT (ENDDATE_CE) sensitivity:

153 [2026-03-04 16:25:36] Patients disenrolled before study ENDDATE: 2610 / 8598 (30.4 %)

154 [2026-03-04 16:25:36] Avg gap (ENDDATE - ENDDATE_CE): 292.8 days

155 [2026-03-04 16:25:36] [D] LOT1 completeness:

156 [2026-03-04 16:25:51] Retriable failure: ! ODBC failed with error 00000 from

157 [Simba][Hardy][AMBIGUOUS_REFERENCE][AMBIGUOUS_REFERENCE][`lb`.`OBS\_END\_DT`,

158 p`.`OBS_END_DT`].

159 X (80) Syntax or semantic analysis error thrown in server while executing query. Error message from server:

160 org.apache.hive.service.cli.HiveSQLException: Error running query: org.apache.spark.sql.AnalysisException: Reference

161 `OBS\_END\_DT` is ambiguous, could be: . SQLSTATE: 42704; 1

162 <SQL>

163 SELECT


```
FROM lot1_base_end lb  
INNER JOIN lot_patient_input p ON lb.PATID = p.PATID  
FROM nanodbc/nanodbc.cpp:1726.  
[2026-03-04 16:26:36] Retrying in 20 s (attempt 4 / 4 )  
[2026-03-04 16:27:11] ODBC failed with error 00000 from \  
[2026-03-04 16:27:11] [AMBIGUOUS_REFERENCE][`lb`.`OBS_END_DT`, `p`.`OBS_END_DT`].  
WARNING: Validation QC suite failed:  
[simba][Hardy][AMBIGUOUS_REFERENCE] and Figures...  
[2026-03-04 16:27:11] Generating descriptive summary and figures...  
=====
```

PART 2 DESCRIPTIVE SUMMARY

5A. MMA_MED (Medication Claims) Summary

Total claims (de-duped) : 358647
Distinct patients: 8528
Distinct medications: 25
Pharmacy claims: 96661
Medical claims: 261986
Date range: 2016-01-05 to 2025-06-30
Avg day supply: 27.6 (median: 28)

Med	Class	Claims	Patients	RX	Medical
BORT	PROTINHIB	142178	6490	900	141278
LENA	IMMUNOMOD	74334	5747	74332	2
DARA	ACD38	80434	4323	974	79460
CYCL	MUSTARD	13647	1702	3088	10559
CARF	PROTINHIB	24193	1127	130	24063
POMA	IMMUNOMOD	10028	1077	10028	0
MELP	MUSTARD	722	466	236	486

```
new 1 - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins
new 2 new 1 new 3
227 MELP MUSTARD 722 466 236 486
228 IXAZ PROTINHIB 5619 463 5619 2215
229 ELDT ASIAME7 2259 164 44 498
230 DOXO TOPOINHIB 504 112 6 1203
231 ISAT ACD38 1209 99 6 0
232 VENE BLC2I 702 91 702 597
233 BEND MUSTARD 604 84 7 0
234 SELI NUCLEAR 191 75 191 0
235 THAL IMMUNOMOD 376 72 376 489
236 ETOP TOPOINHIB 492 66 3 193
237 CISP PLAT 193 42 0 226
238 TALQ BLC2I 226 29 6 372
239 ABCMA ABCMA 378 27 24 24
240 CILT ATCELL 24 18 0 68
241 BELA ABCMA 68 14 0 195
242 ELRA BLC2I 195 10 2 55
243 DOPL TOPOINHIB 57 3 0 3
244 IDEC ATCELL 3 3 11 0
245 PANO HIST 11 3 11
246 Don't know how to automatically pick scale for object of type <integer64>. Defaulting to continuous.
247 [2026-03-04 16:27:54] Figure saved: /mnt/artifacts/results/fig01_mma_patients_by_med.png
248 Don't know how to automatically pick scale for object of type <integer64>. Defaulting to continuous.
249 [2026-03-04 16:27:54] Figure saved: /mnt/artifacts/results/fig02_mma_claims_by_type.png
250
251
252 Day Supply Distribution:
253 Type Min P25 Med P75 Max Mean
254
255 medical 28 28 28 28 28 28.0
256
257 pharmacy 1 28 28 28 100 26.5
258
259
260
Total MAPs: 95650
```

5B. MAP_MED (Medication Available Periods) Summary

Total MAPs: 95650

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File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window

new 2 new 1 new 3

95650
8528

260
261
262
263
264
265
266
267
268
269
270

Total MAPs:
Distinct patients:
Distinct medications:
MAP length (days):
MAPs with discontinuation:

MAPs per patient distribution:
MAPs Patients

Error in sprintf(" %-8d %10s\n", r\$n_maps, format(r\$n_patients, big.mark = ",")) :
invalid format ' %-8d'; use format %f, %e, %g or %a for numeric objects